Branch guidance employees can inspect and question.

Turn a procedure search into a concise answer, with the applicable conditions and visible evidence.

11
supplied procedures

1,388
traceable passages

Local
generative inference

DELIVERED

Working chat
Follow-up questions
Source inspection
Measured evaluation
Six-slide presentation

Decision support over the supplied snapshot.
Employees verify the cited procedure before acting; the assistant does not execute transactions.

Retrieve the evidence before generating the answer.

A transparent RAG pipeline, sized for eleven procedures and a local CPU demonstration.

PREPARE ONCE

Original HTML

11 files · immutable source hashes

Structural parsing

Sections · lists · table conditions

Local indexes

BM25 + 384-dimensional vectors

EACH QUESTION

Question + session context › **Hybrid retrieval + reranking** › **Bounded evidence + local Qwen3** › **Output checks + NLI screening** › **Answer + source passages**

Why RAG: source updates and citations stay explicit. Exact local search avoids unnecessary infrastructure.
Weak retrieval → abstain. Failed answer checks → clearly labelled source excerpts.

Preserve conditions, exceptions and source identity.

Shorter passages reduce competing instructions; structured rows preserve situation-to-action relationships.

Structure-aware preparation

139 HTML tables, including nested tables
~110 narrative words; 20-word overlap for long blocks
Table labels, row conditions and source anchors retained

Retrieval with complementary strengths

BM25: identifiers and exact terms
Vectors: paraphrases and related language
RRF top candidates → cross-encoder reranking
Up to 3 passages / 520 words sent as evidence

EXAMPLE: PR63 / MONTHLY TESTING

Situation and control stay together

Monthly testing → single control
Weekly balancing / random monthly count → dual control

Every passage retains procedure ID, section, publication text, original filename and SHA256.
Changed source files require an index rebuild.

Carry the scenario forward. Retrieve fresh evidence.

Answers, source inspection, clarification, topic changes and honest fallback belong in the same workflow.

DEMONSTRATION SCENARIO / PR63 SOURCE FACTS

What is the coin-counting machine overnight limit?

$5,250 in a single machine. [Source: PR63]

And is monthly testing done in single or dual control?

Single control. Inspect the cited procedure passage.

Useful context, bounded memory

Prior user questions resolve context.
New topics replace old scope.
Previous generated claims are not evidence.

Inspectable outcomes

View sources, reset, export and rate.
Weak evidence: clarify / abstain.
Failed checks: show source excerpts.

Finding evidence and answering correctly are separate tests.

30 procedure questions · 4 two-turn conversations · 6 guardrail cases · AI-authored, source-checked labels.

EVIDENCE HIT IN TOP FIVE



Hybrid MRR@5: 0.928
This measures evidence ranking, not answer accuracy.

GENERATED ANSWER DIAGNOSTICS

73.3% all required fact groups

8 withheld drafts / 35 generated evaluations
Guardrail expected status: 100.0%
CPU p50 / p95: 6.7s / 12.5s

Fact matching and NLI can be wrong.
Review accuracy, relevance, completeness and conciseness independently.

Validate the workflow before an operational pilot.

The prototype demonstrates the approach; independent review determines where it is safe and useful.

Working now

Local generative chat
All 11 sources indexed
Inspectable citations
Follow-ups and topic switches
Measured retrieval alternatives
Visible abstention / fallback

Validate next

Independent SME labels
Accuracy and missing exceptions
Stronger-model comparison
Threshold calibration
Employee task success
Time-to-guidance baseline

Before a pilot

SSO and source entitlements
Authoritative policy versions
Redacted audit / retention
Load tests and monitoring
Escalation and rollback
Required enterprise approvals

Recommendation: a scoped, SME-reviewed validation phase with explicit critical-error and coverage gates.